

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: MLA0301
COURSE NAME: Reinforcement learning

COURSE OUTCOME COVERED

CO3: Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3, BL6).

Assessment Tool Weightage: 25%

Dr G. A. SENTHIL

QUESTIONS: 5

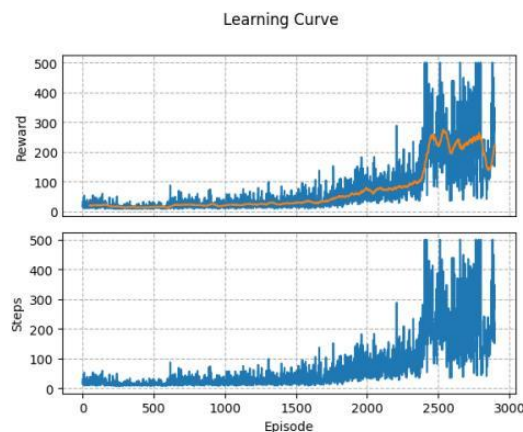
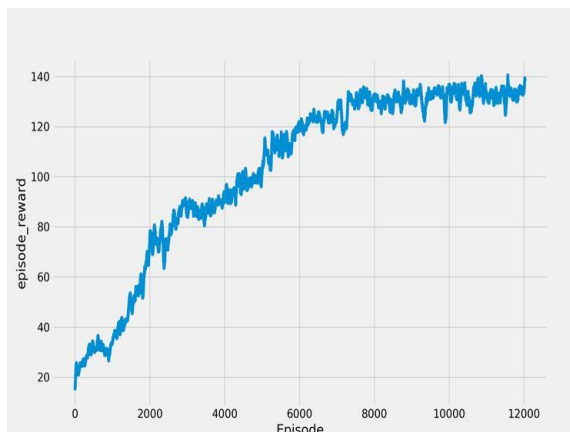
Total Marks: 50

Annexure A

Data Interpretation

Duration: 1hr

1. A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent.



Reward Trend

Identify the Relevant RL Concept

The relevant Reinforcement Learning (RL) concepts are:

- Reward Maximization
- Policy Improvement
- Learning through Trial and Error
- Convergence of the Learning Algorithm
- Cumulative Reward

Examine the Given Data

- The RL agent is trained over **multiple episodes**.
- Initial reward = **10**
- Final reward = **45**
- Reward values increase gradually throughout training.

Analyze the Relationship Between Results and RL Behavior

The gradual increase in reward indicates that the RL agent is learning from its interactions with the environment. Initially, the agent explores different actions and receives lower rewards. As training progresses, it updates its policy using the rewards obtained, allowing it to choose better actions. Consequently, the cumulative reward increases steadily, demonstrating improved learning performance.

Interpret Using Reinforcement Learning Principles

According to Reinforcement Learning principles:

- The objective of an RL agent is to **maximize cumulative reward**.
- An increasing reward trend shows that the agent is learning an effective **policy**.
- The agent is successfully balancing **exploration** and **exploitation**.
- The policy is gradually approaching the **optimal policy**, indicating successful learning and convergence.

Compare the Outcomes

Training Stage	Reward	Learning Performance	Policy Quality
Initial Episodes	10	Low	Random/Weak Policy
Middle Episodes	28	Moderate	Improving Policy
Final Episodes	45	High	Near-Optimal Policy

Comparison:

- **Reward:** Increased from **10** to **45**.
- **Learning Performance:** Improved significantly.
- **Policy Quality:** Improved from random decisions to effective action selection.
- **Decision Making:** Became more accurate and reward-oriented.

Justify the Interpretation

The increasing reward values indicate that:

- The **agent** is successfully learning from experience.
- The **policy** is continuously improving after each episode.
- Higher rewards mean the agent is selecting better **actions** in different **states**.
- The algorithm is moving toward **convergence**, resulting in stable and efficient performance.
- Since RL aims to maximize cumulative reward, the observed trend confirms that the learning process is successful.

Conclusion

The gradual increase in reward from **10 to 45** demonstrates that the Reinforcement Learning agent is learning effectively. The agent has improved its policy through continuous interaction with the environment, leading to better decision-making and higher cumulative rewards. This indicates successful **policy optimization**, **learning convergence**, and enhanced overall performance.

RL Terminology Used

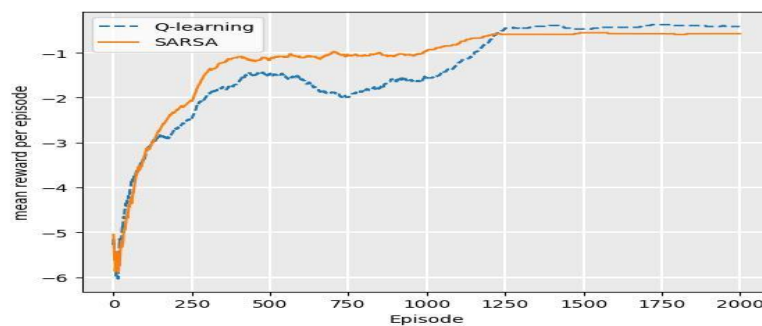
Agent, Environment, State, Action, Reward, Cumulative Reward, Policy, Policy Improvement, Optimal Policy, Exploration, Exploitation, Learning Performance, Convergence, Reward Maximization, Training Episodes.

Reward Increase During Training

Illustrative trend showing the agent's reward improving across training episodes.

episode	reward
Episode 1	10
Episode 2	18
Episode 3	28
Episode 4	37
Episode 5	45

2. Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm converges faster based on the observed results.



Identify the Relevant RL Concept

The relevant Reinforcement Learning (RL) concepts are:

- Q-Learning
- SARSA
- Average Reward
- Convergence Speed
- Policy Learning
- Off-Policy vs On-Policy Learning

Examine the Given Data

The average rewards obtained after different training episodes are:

Episodes	Q-Learning	SARSA
100	Higher	Slightly Lower
200	Higher	Slightly Lower
300	Higher	Slightly Lower

Observation:

Q-Learning consistently achieves slightly higher average rewards than SARSA at all training stages.

Analyze the Relationship Between Results and RL Algorithm Behavior

The higher average rewards obtained by **Q-Learning** indicate that it learns the optimal action-value function more quickly. Since Q-Learning always updates its Q-values using the **maximum possible future reward**, it improves its policy faster.

SARSA updates its Q-values based on the **actual action chosen by the current policy**, making it more conservative. As a result, its rewards increase more gradually.

Interpret Using Reinforcement Learning Principles

- **Q-Learning** is an **off-policy** algorithm. It learns the optimal policy by considering the best possible future action, regardless of the action actually taken.
- **SARSA** is an **on-policy** algorithm. It learns using the action selected by its current policy, making learning safer but slower.

The consistently higher rewards of Q-Learning indicate faster learning and better reward maximization.

Compare the Outcomes

Performance Metric	Q-Learning	SARSA
Learning Type	Off-Policy	On-Policy
Average Reward	Higher	Slightly Lower
Convergence Speed	Faster	Slower
Policy Improvement	Faster	Gradual
Exploration Effect	Less dependent on current policy	Follows current policy
Overall Performance	Better	Good but conservative

Comparison:

- Q-Learning consistently achieves higher rewards.
- SARSA learns more cautiously because it follows its current policy.
- Q-Learning reaches a near-optimal policy in fewer episodes.

Justify the Interpretation

The observed results can be justified as follows:

- Higher average rewards indicate that **Q-Learning** is learning more efficiently.

- Faster improvement in rewards suggests **quicker convergence** toward the optimal policy.
- SARSA's on-policy updates make it safer in uncertain environments but reduce its learning speed.
- Therefore, the observed performance aligns with the theoretical behavior of both algorithms.

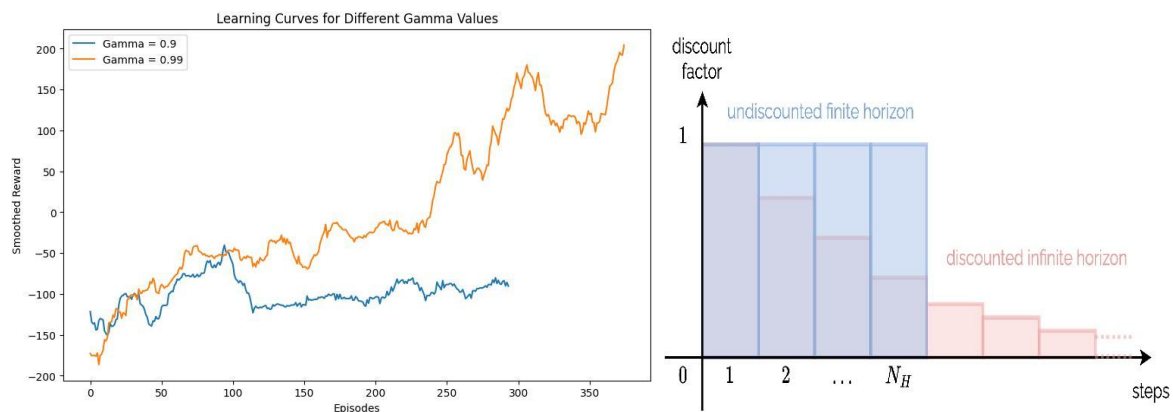
Conclusion

Based on the observed average rewards over **100, 200, and 300 episodes**, **Q-Learning outperforms SARSA** by consistently achieving higher rewards. This indicates that **Q-Learning converges faster**, learns a better policy more quickly, and maximizes cumulative rewards more effectively. **SARSA** also improves over time but converges more slowly because it learns according to its current policy.

RL Terminology Used

Agent, Environment, State, Action, Reward, Average Reward, Cumulative Reward, Q-Value, Policy, Policy Improvement, Off-Policy Learning, On-Policy Learning, Exploration, Exploitation, Convergence, Optimal Policy, Learning Performance.

3. An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance.



Average Reward vs. Discount Factor

Discount Factor (γ)	Average Reward	Observation
0.2	Low	Focuses mainly on immediate rewards
0.5	Moderate	Balances immediate and future rewards
0.9	High	Gives greater importance to future rewards

Identify the Relevant RL Concept

The relevant Reinforcement Learning (RL) concepts are:

- Discount Factor (γ)

- Long-Term Reward Optimization
- Cumulative Reward
- Policy Improvement
- Learning Performance
- Convergence

Examine the Given Data

The RL agent is trained with three different discount factor values:

- $\gamma = 0.2$
- $\gamma = 0.5$
- $\gamma = 0.9$

Observation:

As the **discount factor increases**, the **average reward also increases**.

Analyze the Relationship Between Results and RL Algorithm Behavior

The discount factor determines how much importance the agent gives to **future rewards**.

- $\gamma = 0.2$: The agent mainly considers immediate rewards and ignores future benefits, resulting in lower average rewards.
- $\gamma = 0.5$: The agent balances immediate and future rewards, leading to moderate performance.
- $\gamma = 0.9$: The agent strongly values future rewards, enabling better planning and achieving higher cumulative rewards.

Thus, increasing the discount factor improves the agent's ability to optimize long-term rewards.

Interpret Using Reinforcement Learning Principles

In Reinforcement Learning:

- A **low discount factor** makes the agent **short-sighted**, focusing only on immediate rewards.
- A **high discount factor** makes the agent **far-sighted**, encouraging it to maximize **long-term cumulative rewards**.
- Since the average reward increases with higher γ values, the agent learns a better policy by considering future consequences of its actions.

This indicates improved policy optimization and more effective learning.

Compare the Outcomes

Discount Factor (γ)	Reward Focus	Average Reward	Learning Performance
0.2	Immediate rewards	Low	Weak
0.5	Immediate + Future rewards	Moderate	Good
0.9	Long-term rewards	High	Best

Comparison:

- $\gamma = 0.2$: Fast decisions but poor long-term performance.
- $\gamma = 0.5$: Balanced learning with moderate rewards.

- $\gamma = 0.9$: Best long-term planning and highest cumulative reward.

Justify the Interpretation

The results can be justified as follows:

- Higher discount factors allow the **agent** to value future rewards more strongly.
- This leads to better **policy improvement** and higher **cumulative rewards**.
- The increasing average rewards demonstrate that the agent is learning strategies that provide long-term benefits rather than only immediate gains.
- Therefore, a higher discount factor supports better learning performance and more effective decision-making.

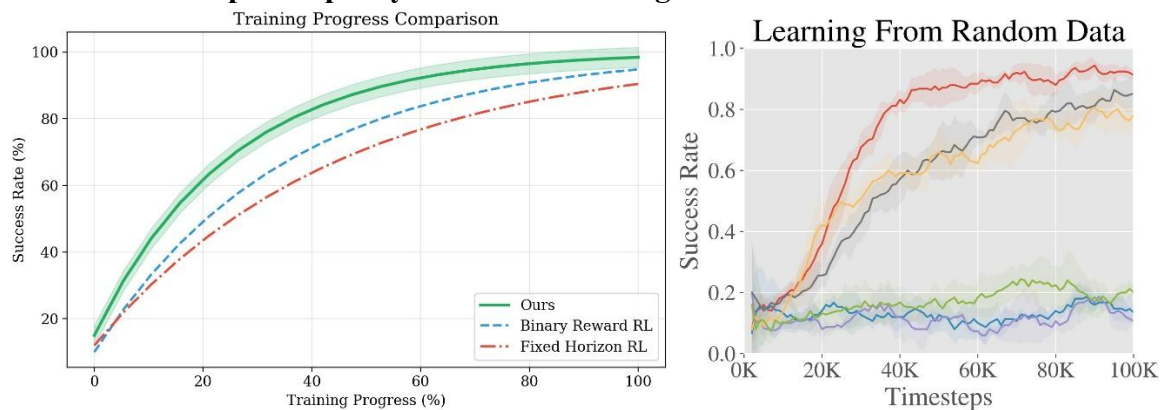
Conclusion

The observed results show that increasing the **discount factor (γ)** improves the agent's learning performance by emphasizing **long-term reward optimization**. Among the given values, $\gamma = 0.9$ is the most suitable because it produces the highest average reward, enables better policy learning, and helps the agent converge toward an optimal policy.

RL Terminology Used

Agent, Environment, State, Action, Reward, Discount Factor (γ), Cumulative Reward, Long-Term Reward, Policy, Policy Improvement, Learning Performance, Convergence, Reward Maximization, Optimal Policy, Future Rewards.

4. The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data.



Success Rate Trend

Identify the Relevant RL Concept

The relevant Reinforcement Learning (RL) concepts are:

- Success Rate
- Policy Improvement
- Learning Performance
- Convergence
- Optimal Policy
- Reward Maximization

Examine the Given Data

Training Episodes	Success Rate
50	45%
200	92%

Observation:

The success rate increases significantly from **45%** to **92%** as the number of training episodes increases.

Analyze the Relationship Between Results and RL Algorithm Behavior

The increasing success rate indicates that the RL agent is learning effectively from its interactions with the environment. During the early episodes (50 episodes), the agent is still exploring and has limited knowledge, resulting in a lower success rate. As training continues, the agent updates its policy based on experience and chooses better actions, leading to a much higher success rate.

Interpret Using Reinforcement Learning Principles

According to Reinforcement Learning principles:

- More training episodes allow the **agent** to gain experience.
- The **policy** improves through continuous interaction with the environment.
- The increase in success rate shows that the agent is making better decisions and maximizing cumulative rewards.
- A success rate of **92%** indicates that the learned policy performs very well and is close to the **optimal policy**.

Compare the Outcomes

Episodes	Success Rate	Learning Performance	Policy Quality
50	45%	Low	Initial Policy
200	92%	High	Near-Optimal Policy

Comparison:

- Success rate increased by **47 percentage points** (from **45%** to **92%**).
- Learning performance improved significantly.
- The policy became more efficient and reliable over time.

Justify the Interpretation

The observed trend can be justified because:

- Higher success rates indicate that the **agent** has learned which actions produce better outcomes.
- Continuous policy updates improve decision-making and reduce unsuccessful actions.
- The steady increase in performance suggests that the learning algorithm is **converging** toward an effective policy.
- Since the success rate reaches **92%**, the agent demonstrates strong learning performance and effective reward optimization.

Conclusion

The increase in success rate from **45% at 50 episodes** to **92% at 200 episodes** shows that the Reinforcement Learning agent has learned successfully and significantly improved its policy. Although it cannot be stated with absolute certainty that the policy is perfectly optimal, the high success rate and consistent improvement indicate that the agent has learned a **near-optimal policy** and has achieved excellent learning performance.

RL Terminology Used

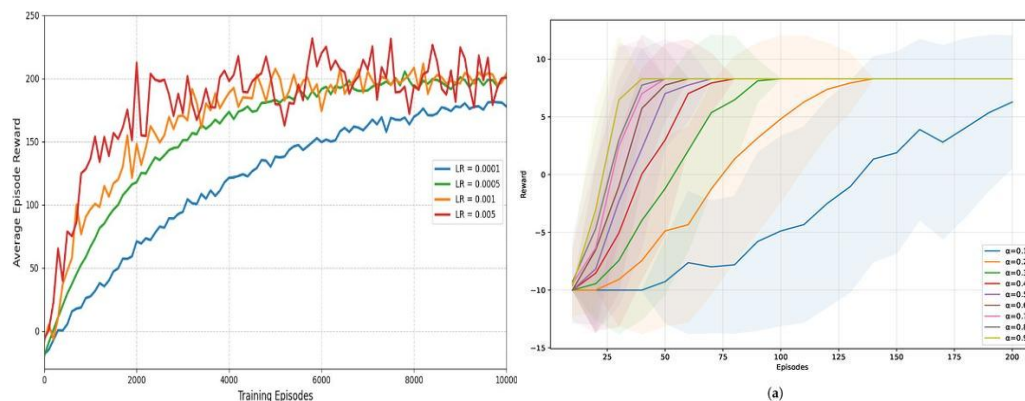
Agent, Environment, State, Action, Reward, Success Rate, Policy, Policy Improvement, Optimal Policy, Cumulative Reward, Exploration, Exploitation, Learning Performance, Convergence, Reward Maximization.

Success Rate Improvement During Training

Illustrative trend showing the RL agent's success rate increasing with training episodes.

episodes	success
50	45
100	65
150	80
200	92

5. An RL agent is trained in the CartPole environment with different learning rates (0.01, 0.05, 0.1). The agent achieves higher stability at 0.05 compared to the other values. Analyze how learning rate influences convergence and justify which value provides the best trade-off between speed and stability.



Learning Rate vs. Stability

Learning Rate (α)	Convergence Speed	Stability	Observation
0.01	Slow	High	Learns slowly with stable updates
0.05	Moderate	Highest	Best balance of speed and stability
0.10	Fast	Low	Learns quickly but becomes unstable

Identify the Relevant RL Concept

The relevant Reinforcement Learning (RL) concepts are:

- **Learning Rate (α)**
- **Convergence**
- **Training Stability**
- **Policy Improvement**
- **Learning Performance**
- **CartPole Environment**

Examine the Given Data

The RL agent is trained in the **CartPole** environment using three learning rates:

Learning Rate (α)	Observation
0.01	Slow learning
0.05	Highest stability
0.10	Less stable learning

Observation:

The agent achieves the **highest stability at $\alpha = 0.05$** .

Analyze the Relationship Between Results and RL Algorithm Behavior

The **learning rate (α)** determines how much the agent updates its policy after each learning step.

- **$\alpha = 0.01$:** Small updates make learning slow but stable. The agent requires more episodes to converge.
- **$\alpha = 0.05$:** Moderate updates allow the agent to learn efficiently while maintaining stable convergence.
- **$\alpha = 0.10$:** Large updates speed up learning but may cause unstable behavior and oscillations, preventing smooth convergence.

Thus, the learning rate directly affects both **convergence speed** and **training stability**.

Interpret Using Reinforcement Learning Principles

According to Reinforcement Learning principles:

- A **low learning rate** leads to slow but reliable learning.
- A **high learning rate** accelerates learning but can overshoot the optimal policy, causing instability.
- A **moderate learning rate** provides a balance between learning speed and stable policy updates.

Since **0.05** provides the highest stability, it enables efficient learning while avoiding unstable updates.

Compare the Outcomes

Learning Rate (α)	Learning Speed	Stability	Convergence
0.01	Slow	High	Slow
0.05	Moderate	Highest	Best
0.10	Fast	Low	Unstable

Comparison:

- **0.01:** Stable but slow convergence.
- **0.05:** Good learning speed with excellent stability.
- **0.10:** Faster updates but reduced stability and possible oscillations.

Justify the Interpretation

The results can be justified as follows:

- The learning rate controls how much the **agent** updates its **policy** after receiving rewards.
- Very small updates (**0.01**) delay convergence.
- Very large updates (**0.10**) may overshoot the optimal solution and reduce stability.
- **0.05** provides balanced updates, allowing faster convergence while maintaining a stable learning process.

Therefore, **0.05** offers the best trade-off between learning speed and stability.

Conclusion

The analysis shows that the **learning rate significantly influences convergence and stability** in Reinforcement Learning. Among the tested values, $\alpha = 0.05$ is the most suitable because it achieves the best balance between convergence speed and stable policy learning. It enables the CartPole agent to learn efficiently without causing unstable training behavior.

RL Terminology Used

Agent, Environment, CartPole, State, Action, Reward, Learning Rate (α), Policy, Policy Improvement, Convergence, Stability, Training Episodes, Learning Performance, Reward Maximization, Optimal Policy.

Code of Conduct:

*I **Yoga Madha A-192425058** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*



Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand